

Artificial Intelligence

For HEDSPI Project

Lecturer 9 – Propositional Logic

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Outline

- What is Logic
- Propositional Logic
 - Syntax
 - Semantic
- Inference in Propositional Logic
 - Forward Chaining
 - Backward Chaining

Knowledge-based Agents

- Know about the world
 - They maintain a collection of facts (sentences) about the world, their **Knowledge Base**, expressed in some **formal language**.
- Reason about the world
 - They are able to derive new facts from those in the KB using some **inference mechanism**.
- Act upon the world
 - They map percepts to actions by **querying** and **updating** the KB.

What is Logic ?

- A **logic** is a triplet $\langle L, S, R \rangle$
 - L, the **language** of the logic, is a class of sentences described by a precise syntax, usually a formal grammar
 - S, the logic's **semantic**, describes the meaning of elements in L
 - R, the logic's **inference system**, consisting of derivation rules over L
- Examples of logics:
 - **Propositional, First Order**, Higher Order, Temporal, Fuzzy, Modal, Linear, ...

Propositional Logic

- Propositional Logic is about **facts** in the world that are either true or false, nothing else
- Propositional variables stand for **basic facts**
- Sentences are made of
 - propositional variables (A,B,...),
 - logical constants (TRUE, FALSE), and
 - logical connectives (not,and,or,..)
- The meaning of sentences ranges over the Boolean values {True, False}
 - Examples: It's sunny, John is married

Language of Propositional Logic

- Symbols
 - Propositional variables: A,B,...,P,Q,...
 - Logical constants: TRUE, FALSE
 - Logical connectives: $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$
- Sentences
 - Each propositional variable is a sentence
 - Each logical constant is a sentence
 - If α and β are sentences then the following are sentences

$$(\alpha), \neg\alpha, \alpha \wedge \beta, \alpha \vee \beta, \alpha \Rightarrow \beta, \alpha \Leftrightarrow \beta$$

Formal Language of Propositional Logic

■ Symbols

- Propositional variables: $A, B, \dots, P, Q, \dots$
- Logical constants: T, F
- Logical connectives: $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$

■ Formal Grammar

- Sentence $\rightarrow$ Asentence | Csentence
- Asentence $\rightarrow$ TRUE | FALSE | A | B | ...
- Csentence $\rightarrow$ (Sentence) | $\neg$ Sentence | Sentence Connective Sentence
- Connective $\rightarrow$ $\neg$ | $\wedge$ | $\vee$ | $\Rightarrow$ | $\Leftrightarrow$

Semantic of Propositional Logic

- The meaning of TRUE is always True, the meaning of FALSE is always False
- The meaning of a propositional variable is either True or False
 - depends on the **interpretation**
 - **assignment of Boolean values to propositional variables**
- The meaning of a sentence is either True or False
 - depends on the interpretation

Semantic of Propositional Logic

■ True table

P	Q	Not P	P and Q	P or Q	P implies Q	P equiv Q
False	False	True	False	False	True	True
False	True	True	False	True	True	False
True	False	False	False	True	False	False
True	True	False	True	True	True	True

$$a \Rightarrow b \Leftrightarrow \neg a \vee b \Leftrightarrow \neg b \Rightarrow \neg a$$

Semantic of Propositional Logic

■ Satisfiability

- A sentence is **satisfiable** if it is true under some interpretation
- Ex: P or H is satisfiable
P and $\neg P$ is unsatisfiable (not satisfiable)

■ Validity

- A sentence is **valid** if it is true in every interpretation
- Ex: $((P \text{ or } H) \text{ and } \neg A) \Rightarrow P$ is valid
P or H is not valid

Semantic of Propositional Logic

■ Entailment

□ Given

- A set of sentences Γ
- A sentence ψ

□ We write

$$\Gamma \models \psi$$

if and only if every interpretation that makes all sentences in Γ true also makes ψ true

- We said that Γ **entails** ψ

Semantic of Propositional Logic

■ Satisfiability vs. Validity vs. Entailment

- ψ is valid iff $\text{True} \models \psi$ (also written $\models \psi$)
- ψ is unsatisfiable iff $\psi \models \text{False}$
- $\Gamma \models \psi$ iff $\Gamma \cup \{\neg\psi\}$ is unsatisfiable

Inference in Propositional Logic

- Forward Chaining
- Backward Chaining

Forward Chaining

- Given a set of rules, i.e. formulae of the form

$$p_1 \wedge p_2 \wedge \dots \wedge p_n \Rightarrow q$$

and a set of known facts, i.e., formulae of the form

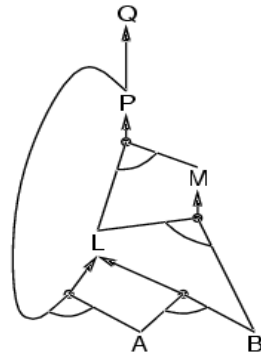
$$q, r, \dots$$

- A new fact p is added
- Find all rules that have p as a premise
- If the other premises are already known to hold then
 - add the consequent to the set of know facts, and
 - trigger further inferences

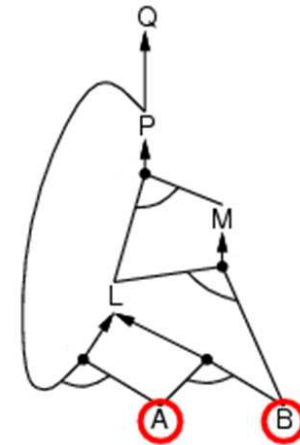
Forward Chaining

■ Example

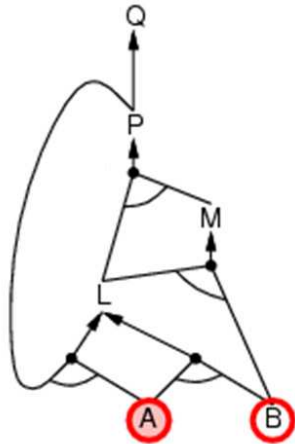
$P \Rightarrow Q$
 $L \wedge M \Rightarrow P$
 $B \wedge L \Rightarrow M$
 $A \wedge P \Rightarrow L$
 $A \wedge B \Rightarrow L$
 A
 B



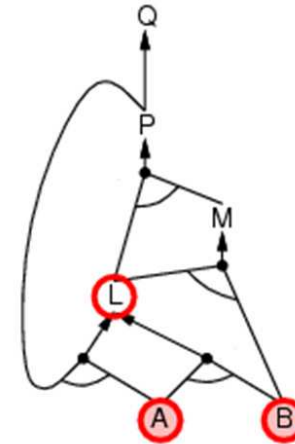
Forward Chaining



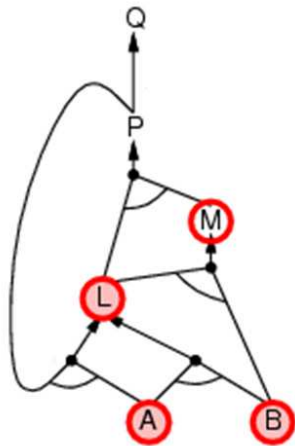
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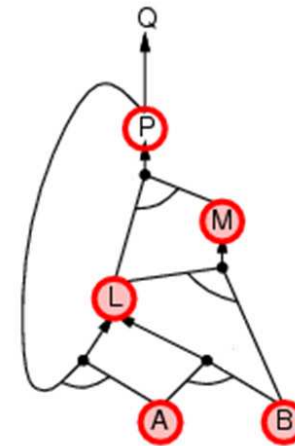
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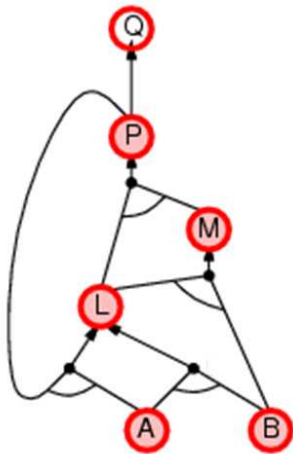
Forward Chaining



Forward Chaining



Forward Chaining



Forward Chaining

- Soundness
 - Yes
- Completeness
 - Yes

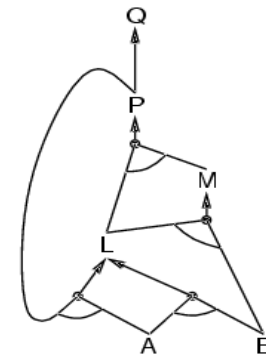
Backward Chaining

- Given a set of rules, and a set of known facts
- We ask whether a fact P is a consequence of the set of rules and the set of known facts
- The procedure check whether P is in the set of known facts
- Otherwise find all rules that have P as a consequent
 - If the premise is a conjunction, then process the conjunction conjunct by conjunct

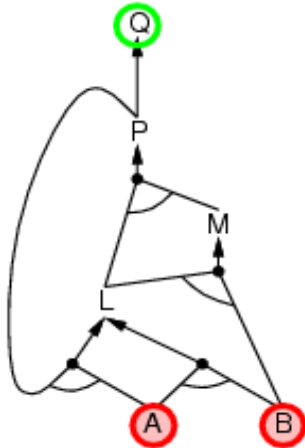
Backward Chaining

■ Example

$P \Rightarrow Q$
 $L \wedge M \Rightarrow P$
 $B \wedge L \Rightarrow M$
 $A \wedge P \Rightarrow L$
 $A \wedge B \Rightarrow L$
 A
 B

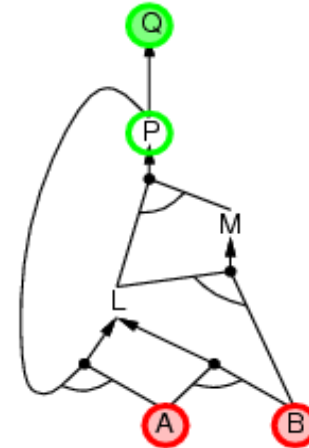


Backward Chaining



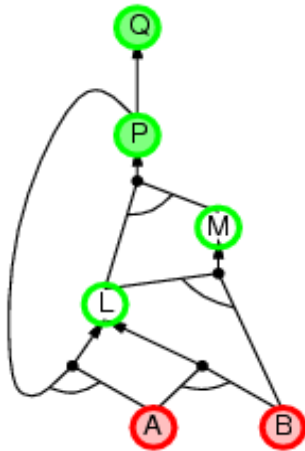
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Backward Chaining



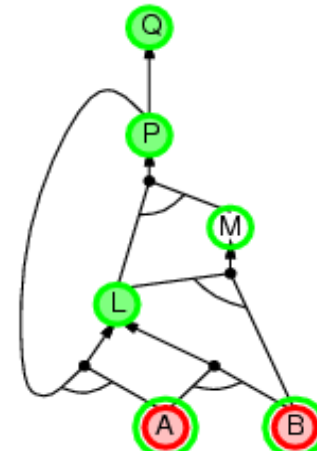
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Backward Chaining



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Backward Chaining



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Backward Chaining

- Soundness
 - Yes
- Completeness
 - Yes

Transformation rules

$$\begin{array}{ll}
 (\alpha \wedge \beta) \equiv (\beta \wedge \alpha) & \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{giao hoán} \\
 (\alpha \vee \beta) \equiv (\beta \vee \alpha) & \\
 ((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma)) & \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{kết hợp} \\
 ((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma)) & \\
 \neg(\neg\alpha) \equiv \alpha & \text{phủ định kép} \\
 (\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha) & \text{tương phản} \\
 (\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta) & \\
 (\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) & \\
 \neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta) & \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{de Morgan} \\
 \neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta) & \\
 (\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) & \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{phân phối} \\
 (\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) &
 \end{array}$$

Transformation rules (con't)

Luật hấp thu:

- $(A \vee (A \wedge B)) \equiv A$
- $(A \wedge (A \vee B)) \equiv A$

Các luật về 0, 1:

- $A \wedge 0 \Leftrightarrow 0$
- $A \vee 0 \Leftrightarrow A$
- $A \vee 1 \Leftrightarrow 1$
- $A \wedge 1 \Leftrightarrow A$
- $\neg 1 \Leftrightarrow 0$
- $\neg 0 \Leftrightarrow 1$

Luật bài trung:

- $\neg A \vee A \Leftrightarrow 1$

Luật mâu thuẫn:

- $\neg A \wedge A \Leftrightarrow 0$

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Transform into CNF

$$B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$$

1. Remove $\Leftrightarrow$, replace $\alpha \Leftrightarrow \beta$ by $(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)$.
 $(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

2. Remove $\Rightarrow$, replace $\alpha \Rightarrow \beta$ by $\neg \alpha \vee \beta$.
 $(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg(P_{1,2} \vee P_{2,1}) \vee B_{1,1})$

3. Move negation inward using the de Morgan's rule :
 $(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge ((\neg P_{1,2} \wedge \neg P_{2,1}) \vee B_{1,1})$

4. Applying the "and" distribution rule :
 $(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg P_{1,2} \vee B_{1,1}) \wedge (\neg P_{2,1} \vee B_{1,1})$

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Example

$$(A \vee B) \rightarrow (C \rightarrow D)$$

1. Remove $\Rightarrow$

$$\neg(A \vee B) \vee (\neg C \vee D)$$

2. Move negation inward

$$(\neg A \wedge \neg B) \vee (\neg C \vee D)$$

3. Distribution

$$(\neg A \vee \neg C \vee D) \wedge (\neg B \vee \neg C \vee D)$$

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Exercises

Transform the following expression into CNF.

1. $P \vee (\neg P \wedge Q \wedge R)$

2. $(\neg P \wedge Q) \vee (P \wedge \neg Q)$

3. $\neg(P \Rightarrow Q) \vee (P \vee Q)$

4. $(P \Rightarrow Q) \Rightarrow R$

5. $(P \Rightarrow (Q \Rightarrow R)) \Rightarrow ((P \wedge S) \Rightarrow R)$

6. $(P \wedge (Q \Rightarrow R)) \Rightarrow S$

7. $P \wedge Q \Rightarrow R \wedge S$

8. $((a \vee b) \wedge c) \rightarrow (c \wedge d)$

Priority: $\neg \wedge \vee \rightarrow \leftrightarrow$

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$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha)$	
$(\alpha \vee \beta) \equiv (\beta \vee \alpha)$	
$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma))$	
$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma))$	
$\neg(\neg\alpha) \equiv \alpha$	
$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha)$	
$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta)$	
$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha))$	
$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta)$	
$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta)$	1. $P \vee (\neg P \wedge Q \wedge R)$
$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma))$	2. $(\neg P \wedge Q) \vee (P \wedge \neg Q)$
$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma))$	3. $\neg(P \Rightarrow Q) \vee (P \vee Q)$
	4. $(P \Rightarrow Q) \Rightarrow R$
	5. $(P \Rightarrow (Q \Rightarrow R)) \Rightarrow ((P \wedge S) \Rightarrow R)$
	6. $(P \wedge (Q \Rightarrow R)) \Rightarrow S$
	7. $P \wedge Q \Rightarrow R \wedge S$